

Analysis General Exam, August 2010

August, 2010

Closed book, closed notes. Please pledge. In each problem, justify all assertions, show calculations, and identify those theorems which you invoke in your arguments.

Problem 1

Let $f(x)$ be a real, normalized function in $L^2(\mathfrak{R})$,

$$\int_{\mathfrak{R}} |f(x)|^2 dx = 1$$

Show that (m is Lebesgue measure):

(a) For $c > 0$,

$$m\{x : |f(x)| \geq c\} \leq \frac{1}{c^2}$$

(b) For $c > 0$ and $r > -1/2$,

$$\left| \int_{[0,c]} x^r f(x) dx \right| \leq \frac{c^{\frac{2r+1}{2}}}{\sqrt{2r+1}}$$

(c) Suppose that additionally,

$$\int_{\mathfrak{R}} x^2 |f(x)|^2 dx < \infty$$

Show that then

$$\int_{\mathfrak{R}} |f(x)| dx < \infty$$

Problem 2

Let $0 < r < 1$ and let

$$\begin{aligned} S_N(x) &= \sum_{n \geq 1}^N \frac{\cos(nx)}{n^{1+r}} \\ S(x) &= \sum_{n \geq 1}^{\infty} \frac{\cos(nx)}{n^{1+r}} \end{aligned}$$

(a) Show that there is a constant c_1 such that

$$|S_N(x) - S_N(y)| \leq c_1 |x - y| N^{1-r}$$

(b) Show that there is another constant c_2 such that

$$|S(x) - S_N(x)| \leq c_2 N^{-r}$$

(c) By suitable choice of N depending on x, y , show that $S(x)$ is Hölder continuous with index r , i.e., there is a finite c_3 such that

$$|S(x) - S(y)| \leq c_3 |x - y|^r$$

Problem 3

Let (X, μ) be a measure space, and let $\{f_n\}$ be a sequence of real-valued integrable functions on X which converges to the function f in an L^1 -sense. Suppose that moreover,

$$\sum_n \int_X |f_n(x) - f(x)| d\mu(x) < \infty$$

Show that $\{f_n\}$ converges pointwise a.e. to f . To do so, consider

$$\mu(\cup_{m \geq N} \{x : |f_m(x) - f(x)| \geq \epsilon\})$$

Problem 4

Let (X, μ) be a finite measure space, and let $\{f_n\}$ be a sequence of real-valued measurable functions converging pointwise a.e. to a measurable function f . We say that the sequence $\{f_n\}$ has uniformly absolutely continuous integrals if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$\int_E |f_n| d\mu < \epsilon$$

for all n whenever E is a measurable set with $\mu(E) < \delta$. Show that in this case, $f_n \rightarrow f$ in the norm of $L^1(\mu)$.

Problem 5

Show that for every $\epsilon > 0$, the function

$$f(z) = \sin z + \frac{1}{z - i}$$

has infinitely many zeros in the set $\{z : |\operatorname{Im} z| < \epsilon\}$.

Problem 6

For n an even integer greater than or equal to 4, compute

$$\int_{-\infty}^{\infty} \frac{x^2}{x^n + 1} dx$$

Show all estimates carefully. Your answer should be a "clearly real" number.

Problem 7

Suppose f is meromorphic (analytic except for poles) in $\mathbb{C}$. Show that if

$$\int_{\gamma} [p(z)]^2 f(z) dz = 0$$

for every polynomial $p(z)$ and every piecewise smooth closed curve γ not passing through a pole of f , then f is entire. Hint: First show f is entire under the assumption $\int_{\gamma} p(z) f(z) dz = 0$ for all such p and γ .

Problem 8

Suppose f is entire and $|f(z)| \leq |z|^2$ for all z . Find all possibilities for f .